Shivam Kachawar

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SUMMARY

- Software Engineer focused on backend development, systems programming, and distributed systems. Hands-on experience building backend applications with Java, Spring Boot, and Kafka, including a job scheduler with asynchronous execution and concurrency control. Also built a Redis-compatible in-memory database in Go and a full-stack URL Shortener. Strong foundation in data structures, algorithms, and system design.

EDUCATION

- **Shri Guru Gobind Singhji Institute of Engineering & Technology, Nanded** *2022 – 2026*
  B.Tech – Computer Science and Engineering *CGPA: 8.76*

EXPERIENCE

- **Algoquant Fintech Ltd** *Jun 2026 – Present*
  Software Engineer (DevOps) *Bangalore, India*
  - Built two internal **Flask web applications**: a Kubernetes job-monitoring dashboard (30s refresh) with REST endpoints exposing job status, logs, and pod events by namespace; and a server inventory dashboard with daily hardware audits and Excel export.
  - Designed and built an automated **data-sync service** in Python, integrating 2–3 exchange contract databases via SQL queries and Cron-based scheduling, eliminating roughly 1 hour of manual reconciliation daily.
  - Deployed and managed a production Kubernetes cluster comprising **23 bare-metal nodes, 1,590 CPU cores, and approximately 10 TiB of RAM**, alongside OpenStack for cloud infrastructure operations and monitoring.
  - Designed and implemented Layer-2 Network Access Control with PacketFence, pfSense, Zentyal LDAP, and RADIUS authentication, including inline traffic interception and captive portal enforcement.
- **Algoquant Fintech Ltd** *Jan 2026 – Jun 2026*
  DevOps Intern *Bangalore, India*
  - Supported enterprise monitoring, infrastructure automation, and Kubernetes/OpenStack administration as part of the DevOps team.

SKILLS

- **Languages:** Java, Go, Python
- **Backend:** Spring Boot, REST APIs, Redis, Apache Kafka, JWT, OAuth 2.0
- **Databases:** PostgreSQL, MySQL, Firebase
- **DevOps & Cloud:** Linux, Kubernetes, OpenStack, Docker, Git
- **CS Fundamentals:** Data Structures & Algorithms, System Design (LLD & HLD), OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

- **Job Scheduler | Java, Spring Boot, PostgreSQL, Kafka, React** [GitHub - Backend] [GitHub - Frontend]
  - Built a job scheduler supporting **3 scheduling modes** (one-time, interval, cron) using an in-memory priority queue and PostgreSQL-backed state for restart recovery.
  - Implemented asynchronous execution with **Apache Kafka** and the **Transactional Outbox Pattern**, ensuring atomic event persistence and at-least-once delivery.
  - Prevented duplicate claims through atomic database updates; implemented configurable retries with exponential backoff and per-attempt tracking.
  - Benchmarked **10,000 executions** across **1–50 concurrent workers**, averaging **14.9K claims/sec** at 10 workers across 3 local runs; built a **React dashboard** and wrote **26 automated tests** using JUnit, Mockito, and Testcontainers, including PostgreSQL concurrency tests.
- **Sniply - URL Shortener** [Live]
  - Developed a full-stack URL Shortener platform using **Spring Boot**, **React**, **PostgreSQL**, and **Redis**, implementing **Base62 encoding** for compact URL generation, custom aliases, QR code generation, click analytics, and link expiration.
  - Implemented secure authentication using **JWT** and **Google OAuth 2.0**, enabling seamless account linking, role-based access control, and admin functionalities.
  - Implemented **Redis caching** using the **Cache-Aside** pattern with cache invalidation and expiry management, reducing cached-link redirection latency from 200ms to 40–50ms (a 75% improvement).
  - Deployed the application on cloud infrastructure using **Vercel**, **Render**, **Neon PostgreSQL**, and **Upstash Redis Cloud**, enabling scalable access and real-time URL tracking.
- **In-Memory Cache Like Redis | Go** [Github]
  - Built a **Redis-compatible in-memory database** from scratch in **Go**, implementing a **single-threaded, event-driven architecture** using **Linux epoll**, non-blocking sockets, and persistent TCP connections.
  - Implemented the **Redis Serialization Protocol (RESP2)** with incremental parsing, fragmented packet handling, per-client input buffers, and support for command pipelining over a single TCP stream.
  - Developed **50+ Redis-compatible commands** across Strings, Lists, Hashes, Sets, and Sorted Sets using a custom **Skip List**; implemented **AOF persistence**, **Transactions (MULTI/EXEC/DISCARD)**, **Pub/Sub**, and **LRU** cache eviction.
  - Benchmarked the server using **redis-benchmark**, achieving approximately **174K GET requests/sec** and **65K SET requests/sec** with **100K requests** and **50 concurrent clients**.

CODING & ACHIEVEMENTS

- 1st Runner-Up – HackFusion (National Hackathon) - Paperless College Management System *Feb 2025*
- 1st Runner-Up – HackSpectra (National Hackathon) - Resume Tailoring & Job Matching Platform *Apr 2025*
- **1150+ DSA problems** on GeeksForGeeks, LeetCode, takeUForward+